

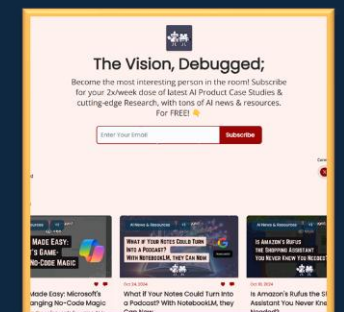
Building AI Agents with AutoGen

Workshop by
Tezan Sahu



A little about me...

- **Microsoft IDC** (2021 – Present)
 - Software Engineer II (2024 – Present) [Enterprise Copilot Extensibility]
 - Data & Applied Scientist II (2023 - 2024) [Bing Autosuggest]
 - Data & Applied Scientist (2021 - 2023) [Bing Autosuggest]
- **IIT Bombay** (2017 – 21)
 - Major in Mechanical Engineering (9.90 CPI)
 - Minor in Computer Science & Engineering
 - Department Rank 1 | Institute Rank 2
- Amazon #1 Best Selling Author of “Beyond Code”
- Co-Author of AI Newsletter “The Vision, Debugged”
- Holder of 2 US Patents
- **AI & Data Science Mentor @Topmate** (2023 – Present)



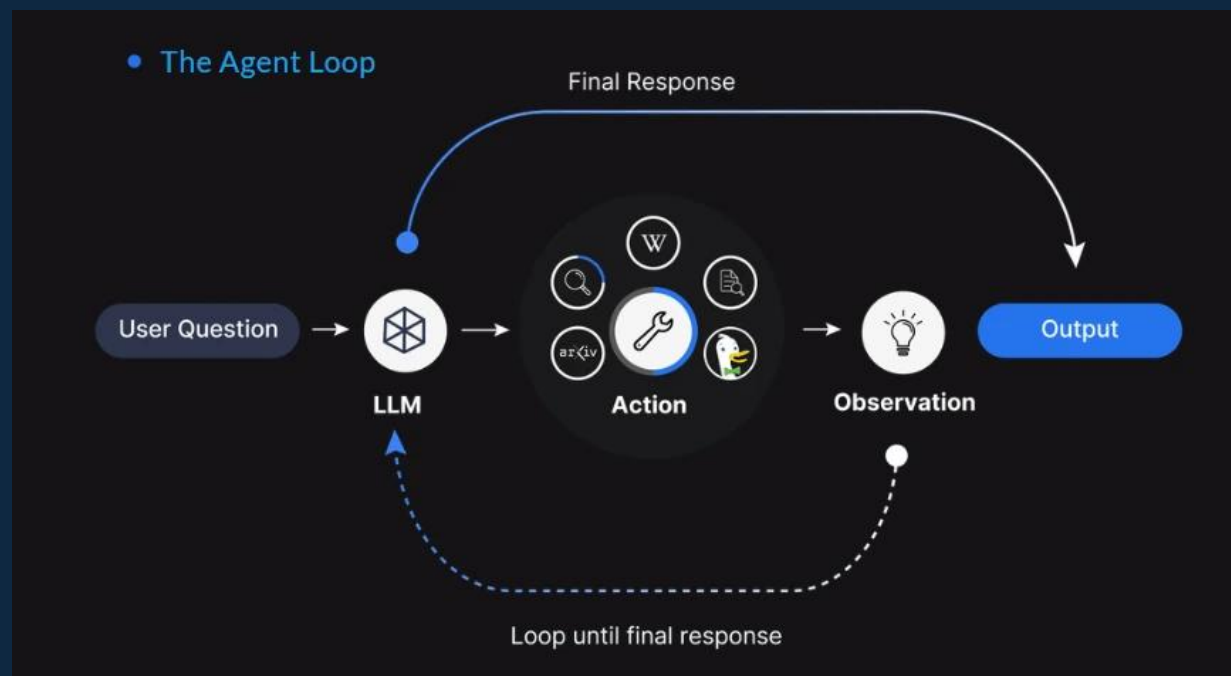
Agenda

- **Introduction to AI Agents**
- **AI Agent Fundamentals**
- **State of AI Agents**
- **Agentic Design Patterns**
- **Introduction to Microsoft's AutoGen**
- **[Hands-On] Building Single-Agent Systems**
- **[Hands-On] Building Multi-Agent Systems**
- **Taking AI Agents to Production**
- **Quiz Time**
- **Q&A**

Introduction to AI Agents



What's an AI Agent?



An AI agent is a **system** or program capable of **autonomously performing tasks** on behalf of a user or another system by **designing its workflow** and **utilizing available tools**

What Makes Agents So Smart?

- **Autonomy:** Operates without constant human oversight
- **Goal-Orientation:** Works towards achieving defined objectives
- **Context-Awareness:** Adapts to changes in the environment or requirements
- **Interactivity:** Communicates with humans and other systems seamlessly



From Clunky Bots to Smart Sidekicks



AI Agent Fundamentals

LLMs – The Brains Behind Agents

Large Language Models (LLMs) are AI models trained on massive datasets to understand and generate human-like text

Core Capabilities

- ✓ **Language understanding**
- ✓ **Text generation**
- ✓ **Summarization, translation, and Q&A**
- ✓ **Few-shot and zero-shot learning**

How do they work?

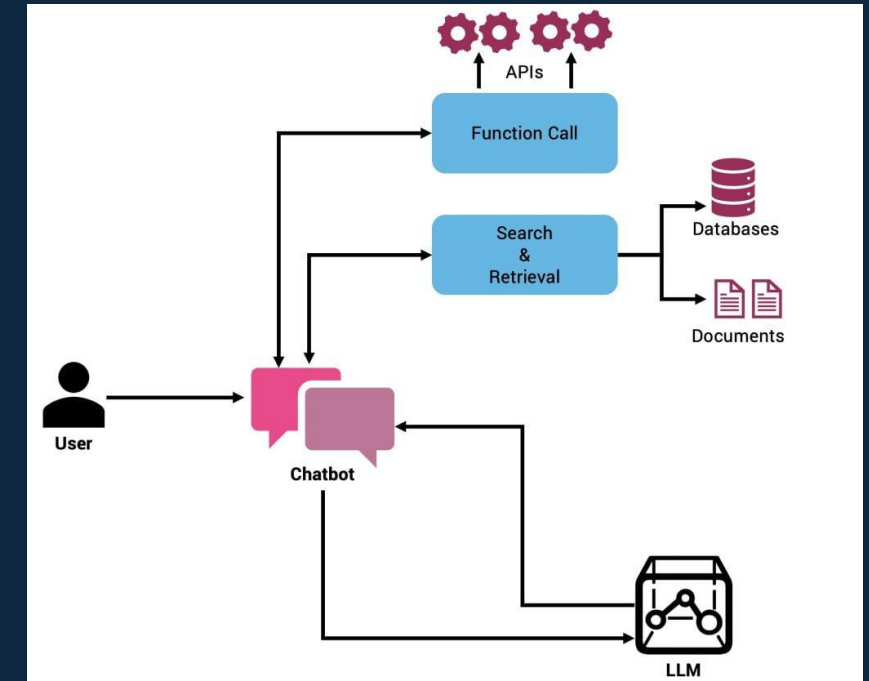
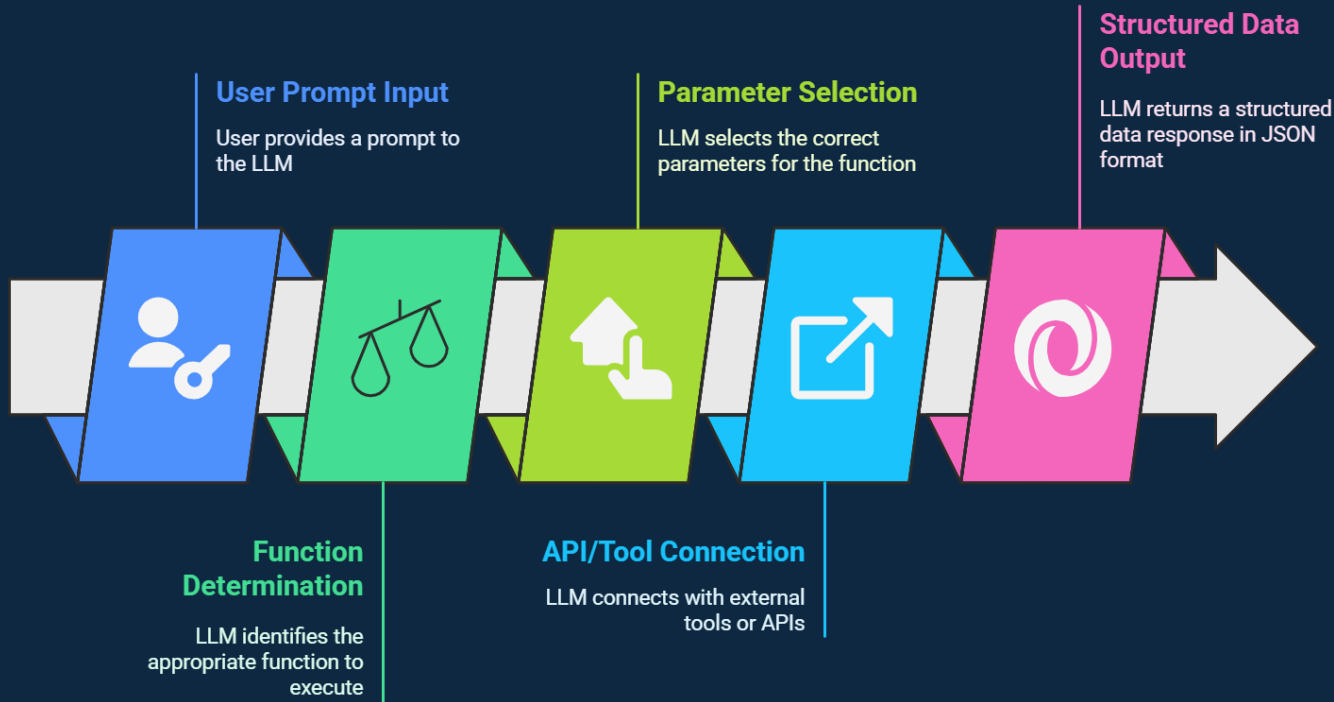
- ▶ **Token-based processing using transformer architecture**
- ▶ **Predict next token based on context**
- ▶ **Fine-tuning and prompt engineering enhance specificity**

Limitations

- ✗ **Lacks real-time knowledge (cutoff dates)**
- ✗ **Prone to hallucination (confident but incorrect)**
- ✗ **No direct interaction with external systems**

Function Calling in LLMs

LLM's ability to determine, from a user prompt, the **appropriate function to execute** from available options and the **correct parameters** to pass to that function



Automation vs Workflows vs Agents

	Automation 	AI workflow 	AI agent 
Definition	A program that executes predefined, rule-based tasks automatically	A program that calls an LLM via API for one or more steps	A program designed to perform non-deterministic tasks autonomously
Core foundations	1011 0110 Boolean logic	1011 0110 Boolean logic ≍ Fuzzy logic	≍ Fuzzy logic ↻ Autonomy
Tasks	Deterministic, predefined tasks	Deterministic tasks requiring flexibility	Non-deterministic, adaptive tasks
Strengths	• Delivers reliable outcomes • Fast to execute	• Better handling of complex rules • Great for pattern recognition	• Highly adaptive to new variables • Simulates human-like behavior and reasoning
Weaknesses	• Limited to tasks explicitly programmed • Cannot adapt to new scenarios • Struggles with complexity	• Requires data to train models effectively • Harder to debug and interpret	• Less reliable, may produce unpredictable undesired outcomes • Slower to execute
Example	Send a Slack notification every time a new lead signs up on our website	Analyze, score and route every website inbound lead using ChatGPT	Perform a full internet search on every inbound lead and update infos

Source: Tyler McGregory via LinkedIn

Automation vs Workflows vs Agents

Which solution fits my business?

Automations

Suitable for repetitive tasks with predefined rules and limited flexibility.

AI Workflows

Ideal for complex processes requiring dynamic decision-making and scalability.

AI Agents

Best for autonomous tasks that require real-time adaptation and learning.



State of AI Agents



What Can AI Agents Do Today?

51.1%

**Companies have
Agents in production**

90%

**Respondents from
Non-Tech Companies
have or plan to have
Agents in production**

In your opinion, which tasks are agents best suited to perform today?



AI Agent Capabilities

Human-like Reasoning

AI agents trace and revise decisions based on new information.

Managing Multistep Tasks

AI agents handle complex tasks with deeper reasoning and context management.

Task Routing & Collaboration

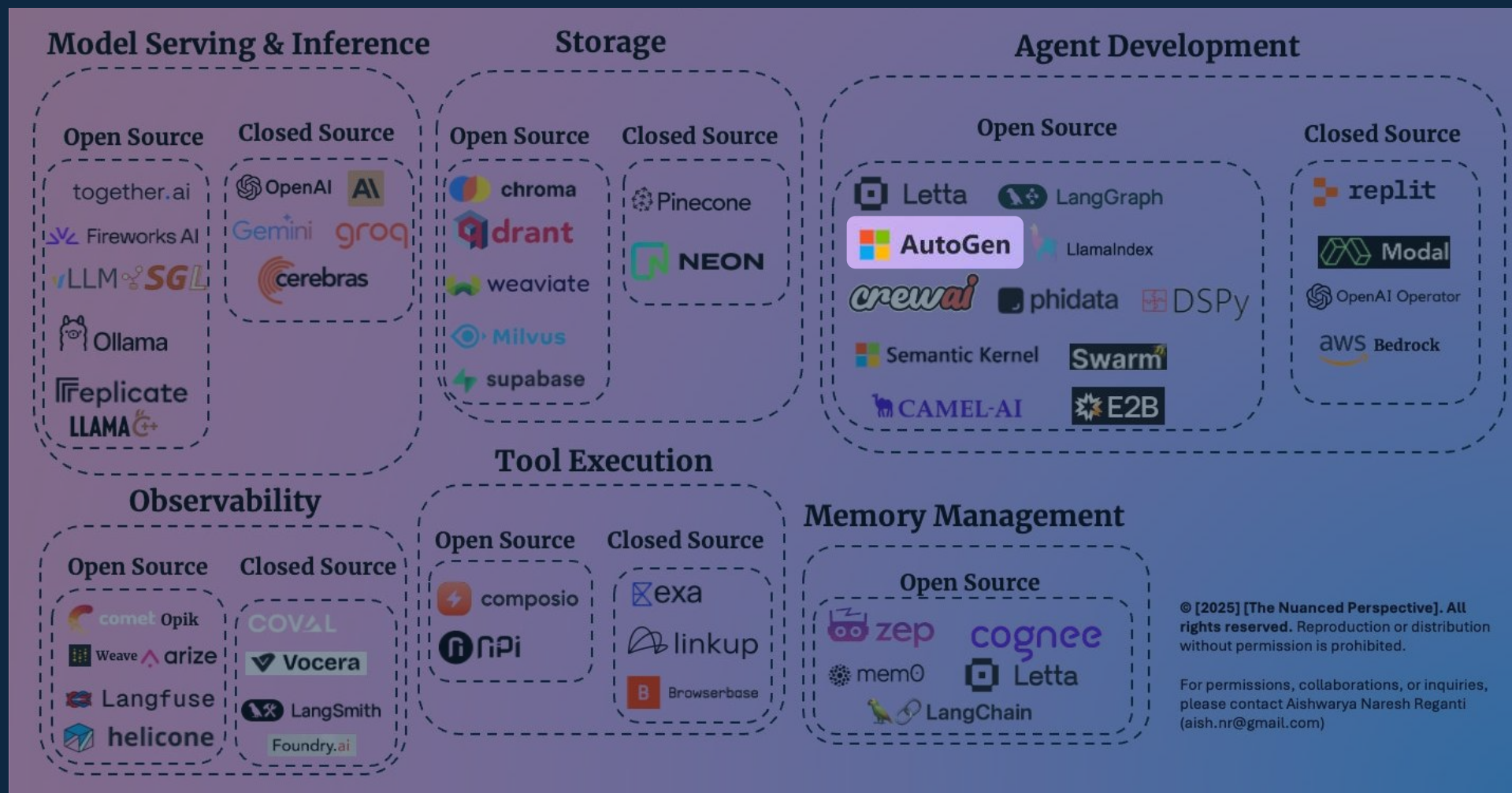
AI agents ensure efficient task routing and collaboration in multi-agent systems.

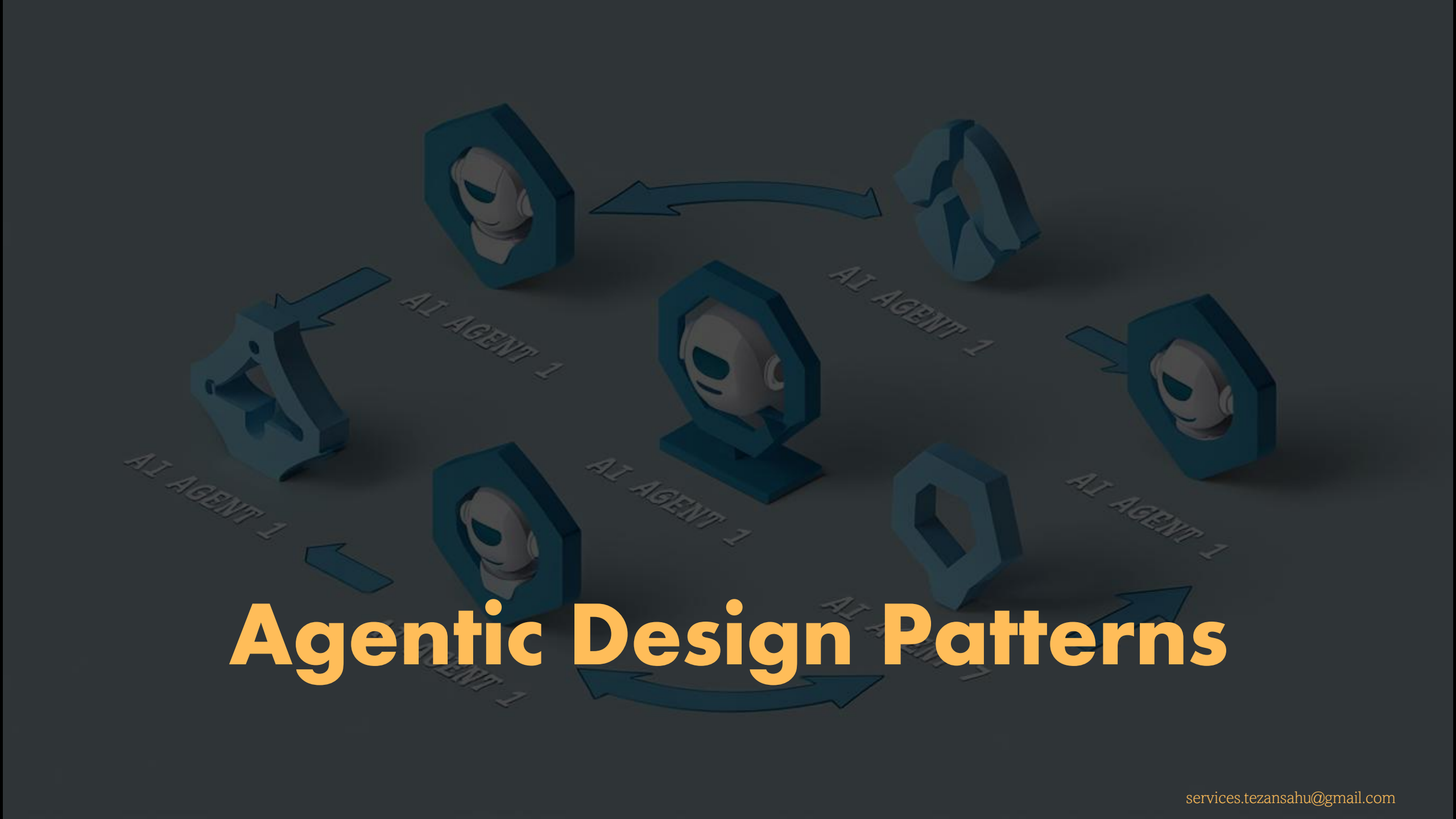
Automating Repetitive Tasks

AI agents automate administrative tasks, freeing time for creative problem-solving.

[Ref] State of AI Agents Report - LangChain

The AI Agent Stack – Jan 2025





Agentic Design Patterns

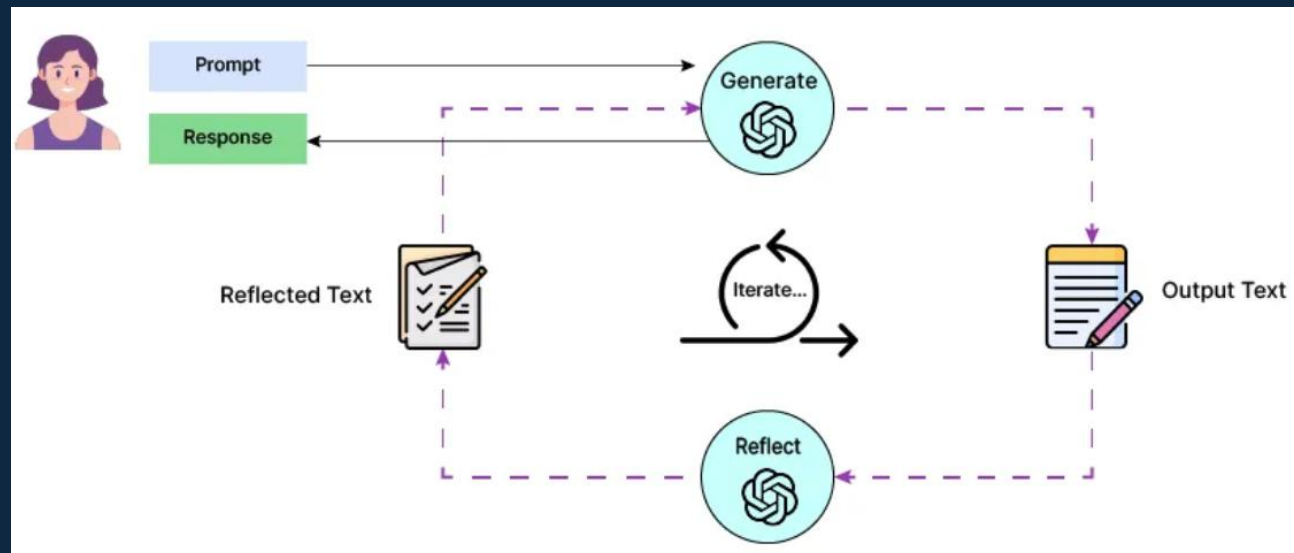
Reflection Pattern – Teach Your Agent to Self-Reflect

- **Key Features:**

- **Self-Evaluation:** Agents assess their performance using metrics or goals
- **Feedback Loops:** They adjust their strategies based on the evaluation

- **Scenario:**

- A recommendation agent reflects on user feedback to improve its suggestions



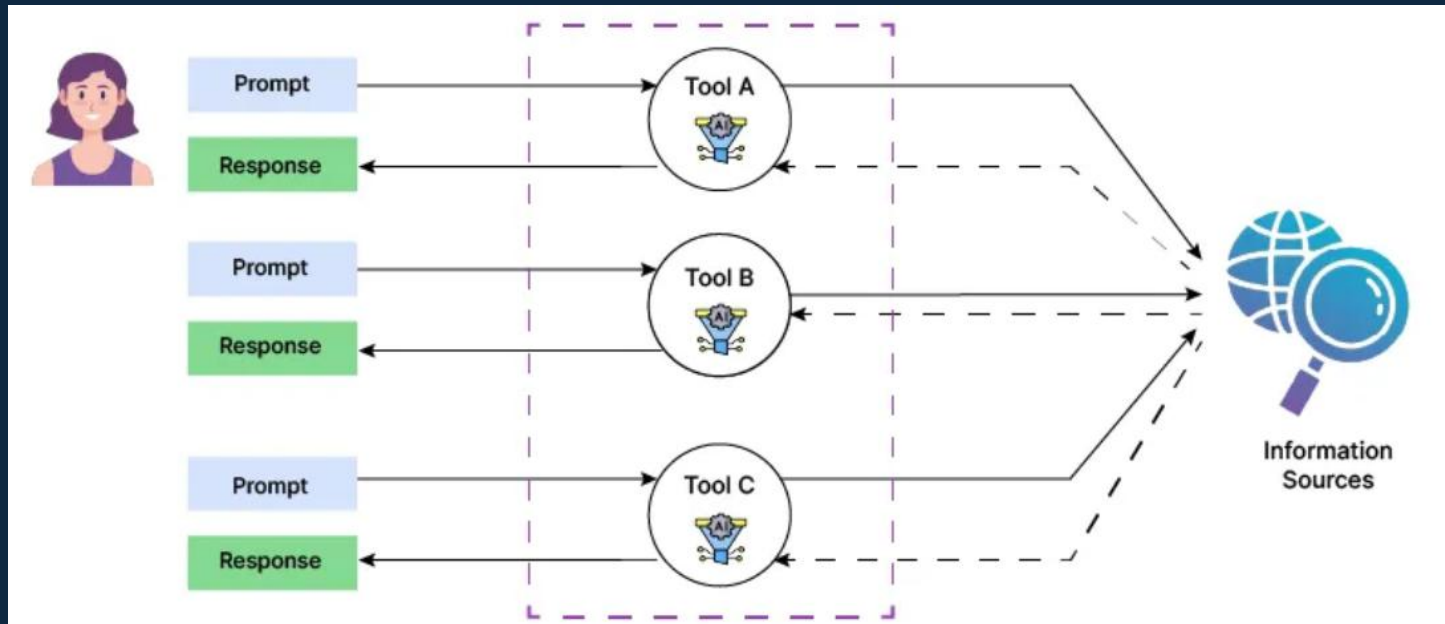
Tool-Use Pattern – Handing AI Its Toolkit

- **Key Features:**

- **Specialized Functionality:** Agents call tools for tasks like data retrieval, transformation, or complex computations
- **Interoperability:** Seamlessly integrates tools into workflows

- **Scenario:**

- An AI agent uses an external API to fetch stock prices and another tool for sentiment analysis



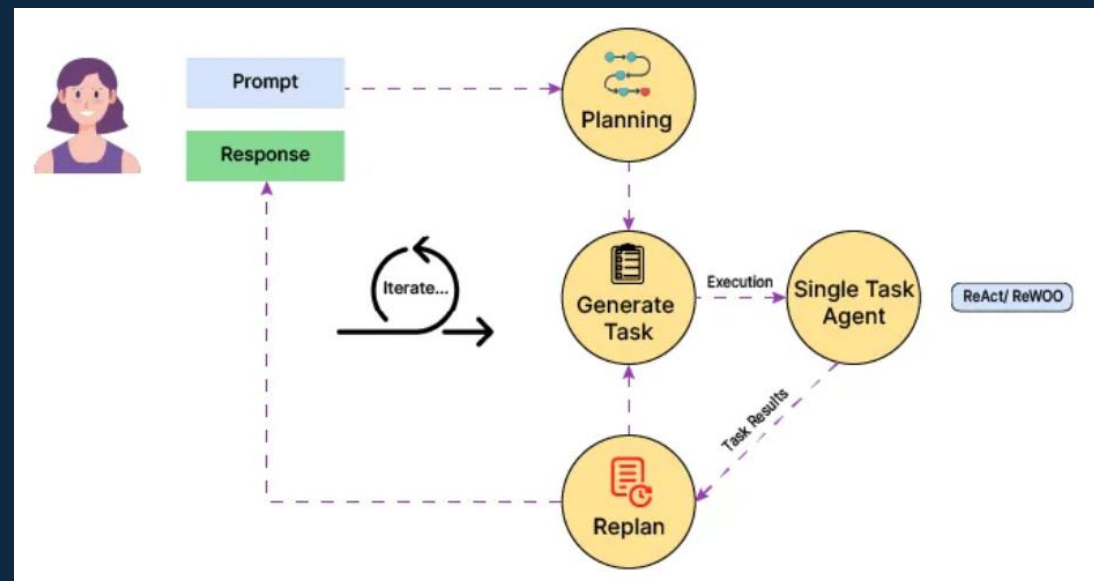
Planning Pattern – Making AI Think Before It Acts

- **Key Features:**

- **Dynamic Goal Alignment:** Plans are generated in real-time based on inputs or changes
- **Sequential Execution:** Agents execute plans step-by-step, ensuring dependencies are managed

- **Scenario:**

- A logistics agent plans delivery routes by optimizing for distance, fuel cost, and traffic conditions



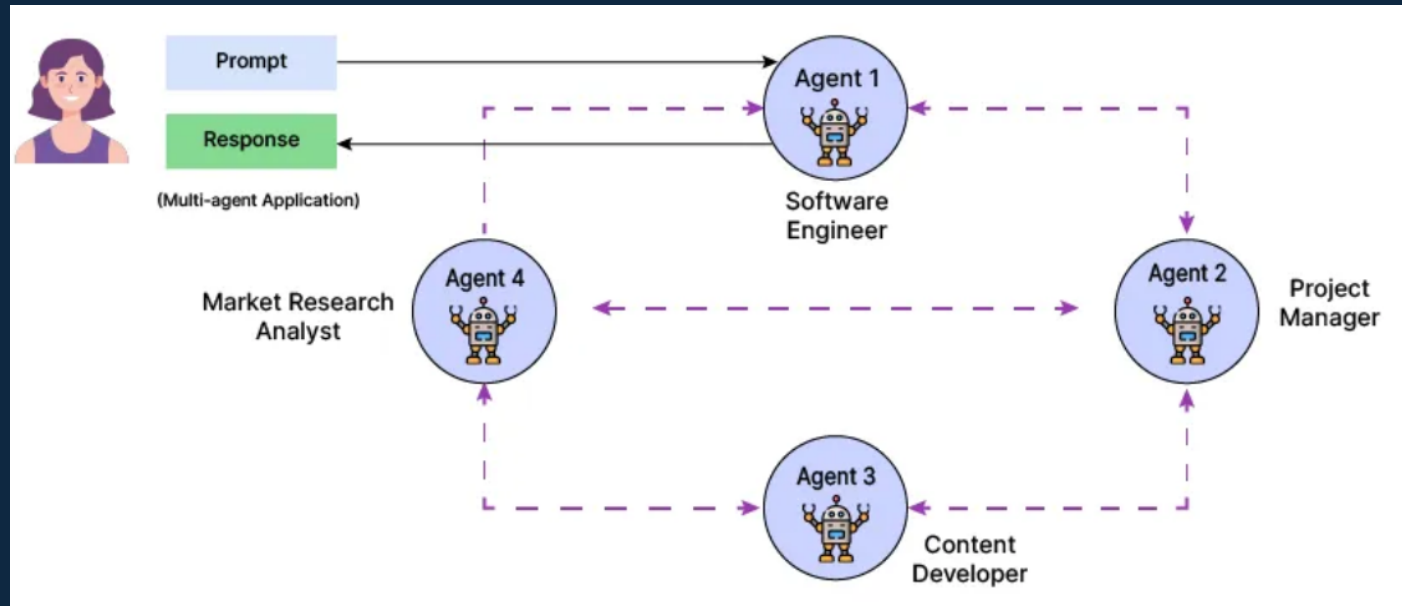
Multi-Agent Pattern – Teamwork Makes Dream Work

- **Key Features:**

- **Division of Labor:** Agents handle different aspects of a task
- **Coordination:** A central orchestrator ensures agents align their actions

- **Scenario:**

- A marketing workflow involves one agent for content generation, another for scheduling, and a third for analytics





Introduction to Microsoft's AutoGen

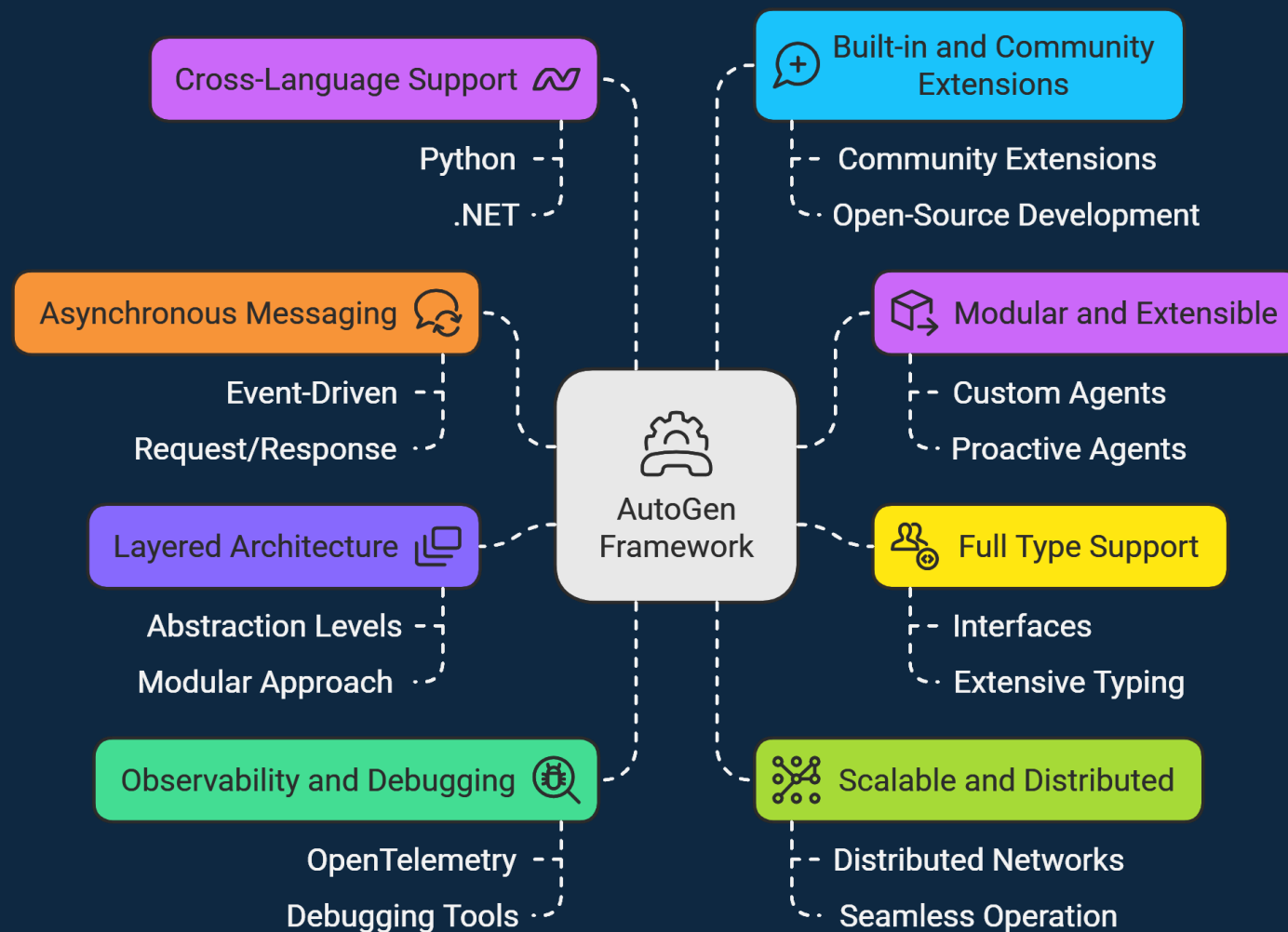
Fundamentals of AutoGen

Open-Source framework for **building AI agents** & facilitating **cooperation** among multiple agents to **solve tasks**

AutoGen Versions:

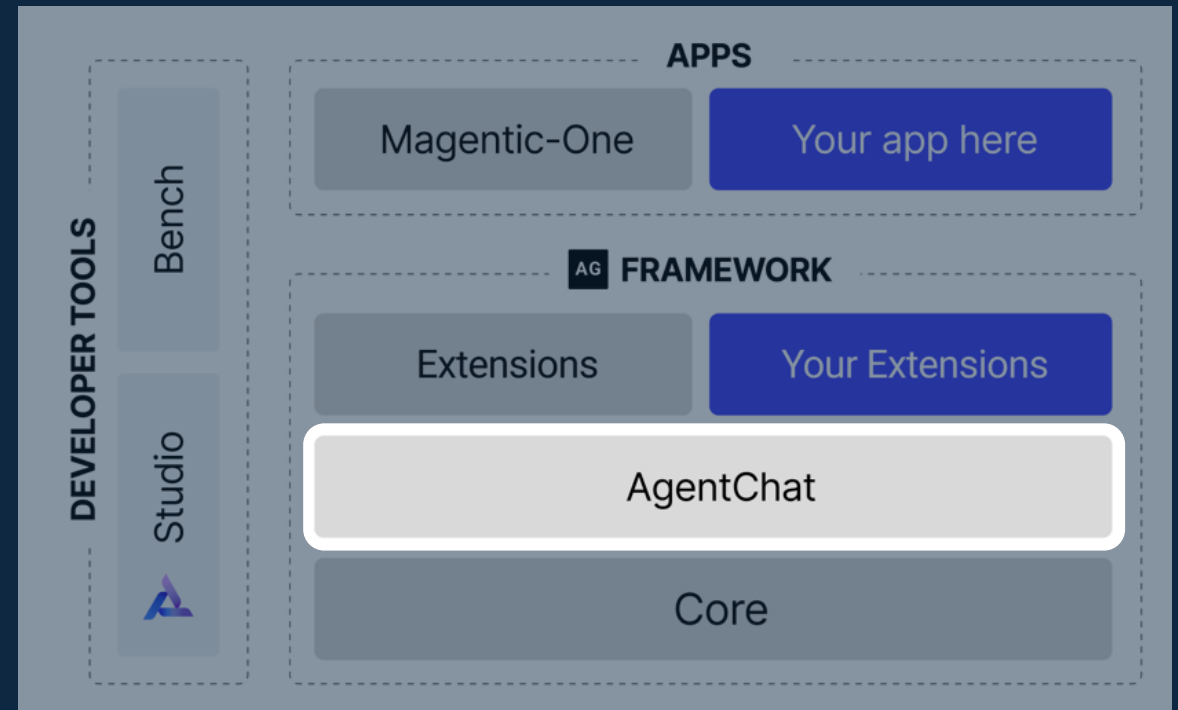
- 0.2 (Deprecated)
- 0.4 (Stable)

- [GitHub Repository](#)
- [Documentation](#)



AutoGen v0.4 Ecosystem

- **Framework**
 - **Core API:** The foundation layer offering a scalable, event-driven actor framework for creating agentic workflows
 - **AgentChat API:** Built on Core, offering a task-driven, high-level framework for building interactive agentic applications
- **Developer Tools**
 - **AutoGen Studio:** Low-code interface that enables rapid prototyping of AI agents
- **Apps**
 - **Magentic-One:** A new generalist multi-agent application to solve open-ended web and file-based tasks across various domains



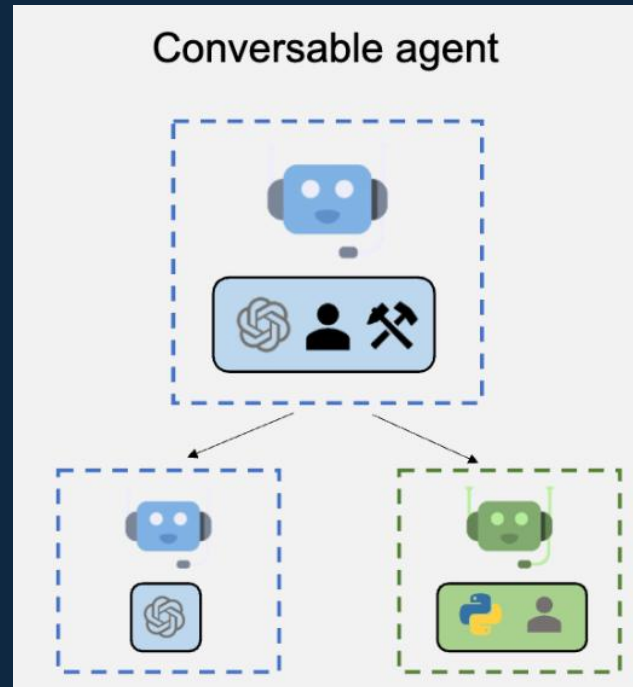
Source: Microsoft DevBlogs

AgentChat Fundamentals – Agents

AssistantAgent

This is a general-purpose agent powered by an LLM.

It can engage in natural language conversations, generate plans, and provide instructions.



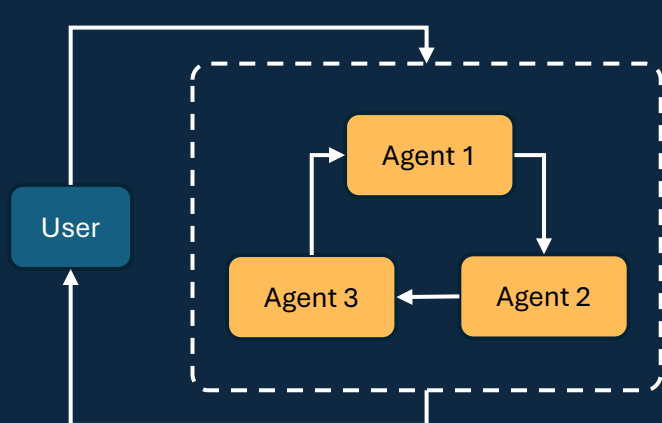
UserProxyAgent

This is a special agent that acts as a proxy for the end-user.

It can relay information between the user and other agents

AgentChat Fundamentals – Teams

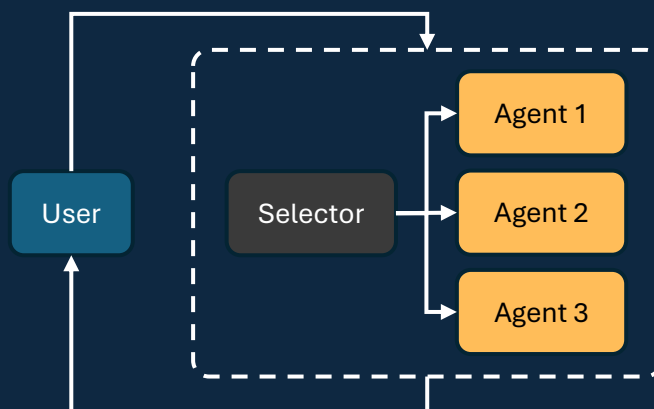
Centralized



Round Robin Group Chat

All agents share the same context and take turns responding in a round-robin fashion.

Each agent, during its turn, broadcasts its response to all other agents

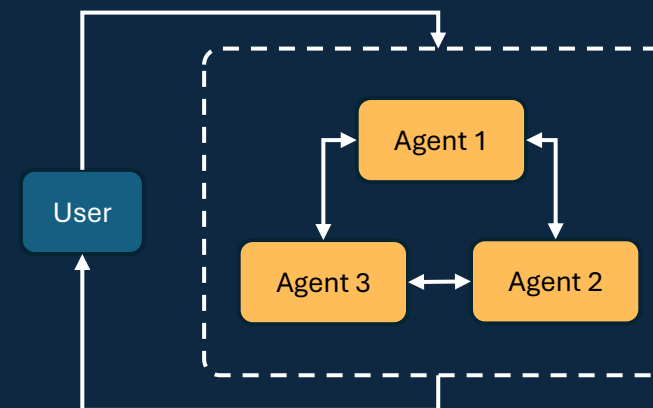


Selector Group Chat

Participants take turns broadcasting messages to all other members.

An LLM selects the next speaker based on the shared context, enabling dynamic, context-aware collaboration

Decentralized



Swarm

A team in which agents can “hand off” tasks to other agents based on their capabilities.

Agents make local decisions about task planning, rather than relying on a central orchestrator

AgentChat Fundamentals – Messages

Agent-Agent Messages

Includes message types for agent-to-agent communication

TextMessage

MultiModalMessage

HandoffMessage

Internal Events

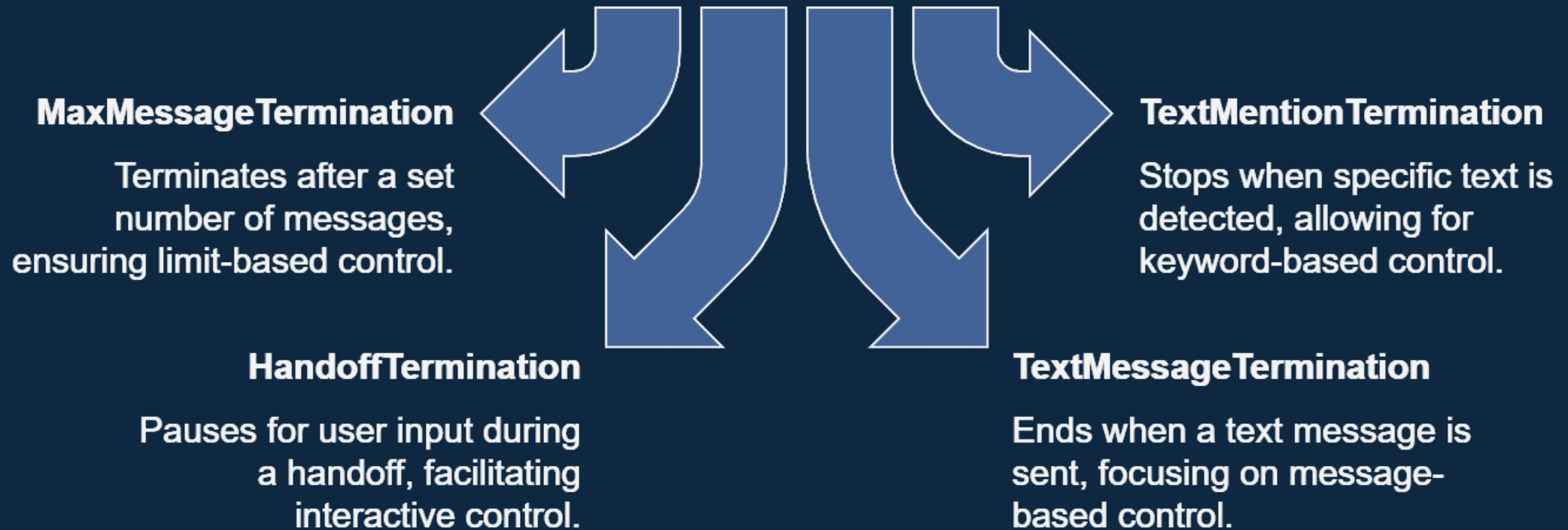
Includes messages used to communicate events and information on actions within the agent itself

ToolCallRequestEvent

ToolCallExecutionEvent

AgentChat Fundamentals – Termination

Common Termination Conditions



Time for some Live Action!



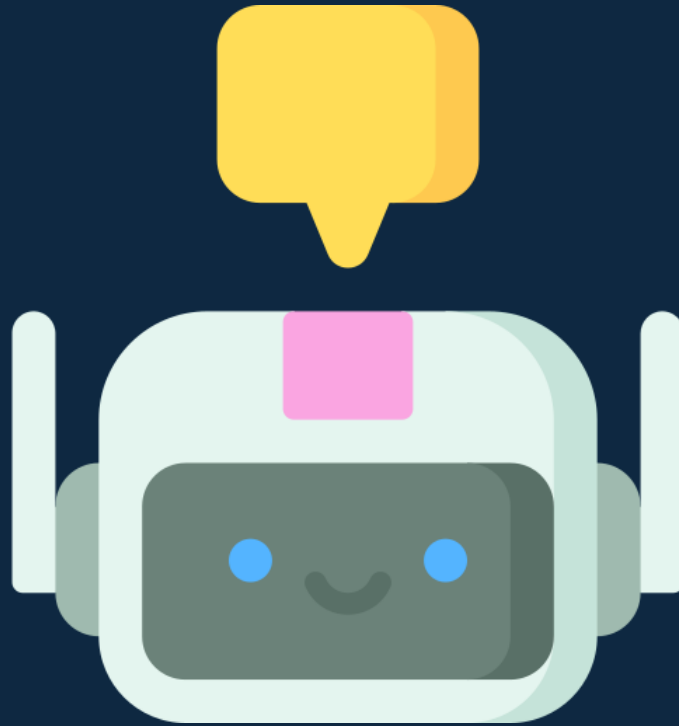
Let's Get Your Environment Ready...

- **Create Virtual Environment**
- **Install all required packages**
- **Set up GitHub Personal Access Token (PAT)**
 - **Access LLMs (GPT-4o, GPT-4o-Mini, etc.) for FREE**
- **Get API Keys for Tools**
 - **Serper**
 - **Firecrawl**
 - **Typefully**

A futuristic robot with a white and grey metallic body is seated at a desk in a dimly lit office. The robot is facing left, looking at several computer monitors that display lines of code. The desk is cluttered with various items, including a coffee cup, a water bottle, and some papers. The overall atmosphere is high-tech and professional.

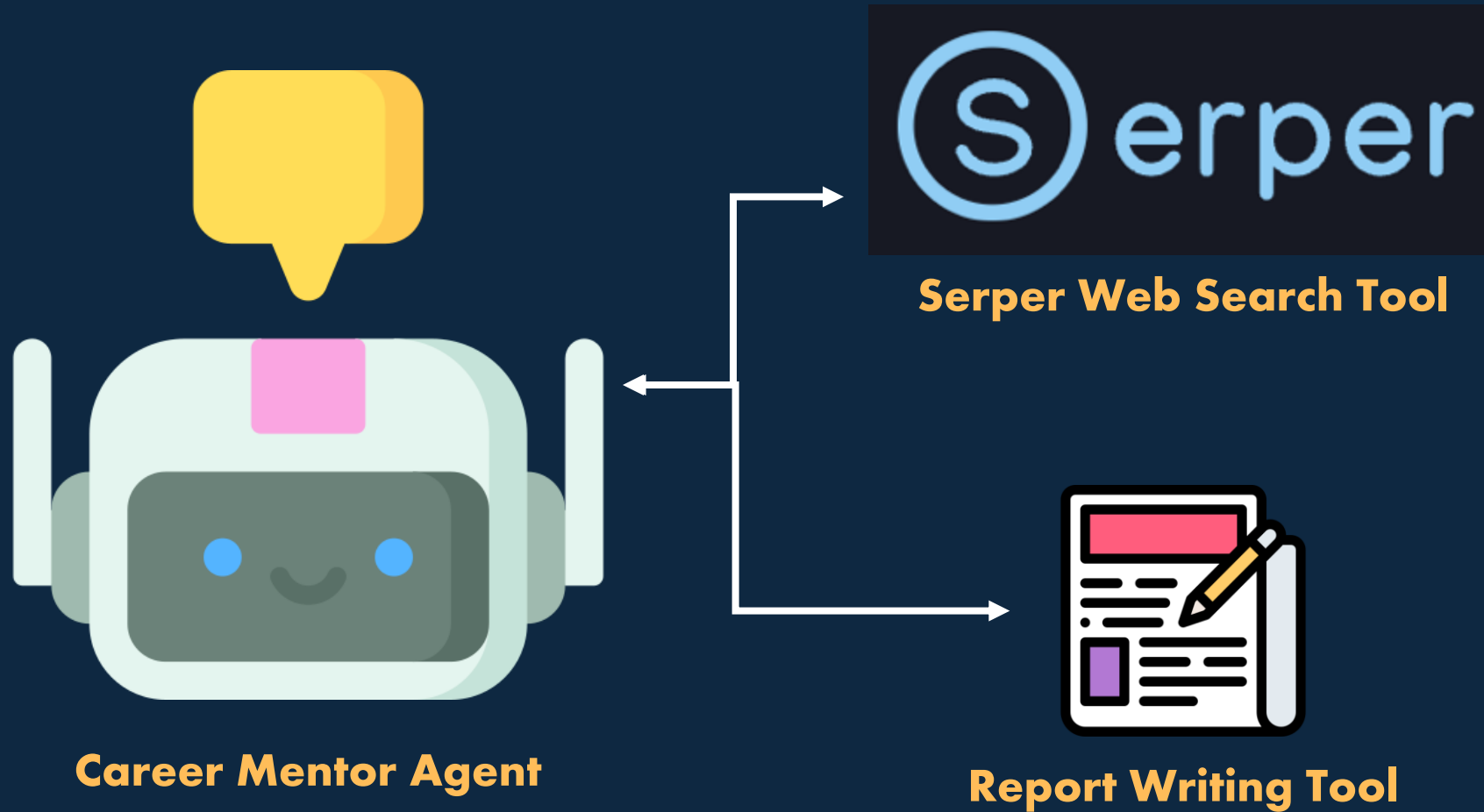
Building Single-Agent Systems

Basic Single Agent – Instructions Only



Career Mentor Agent

Single Agent with Tools

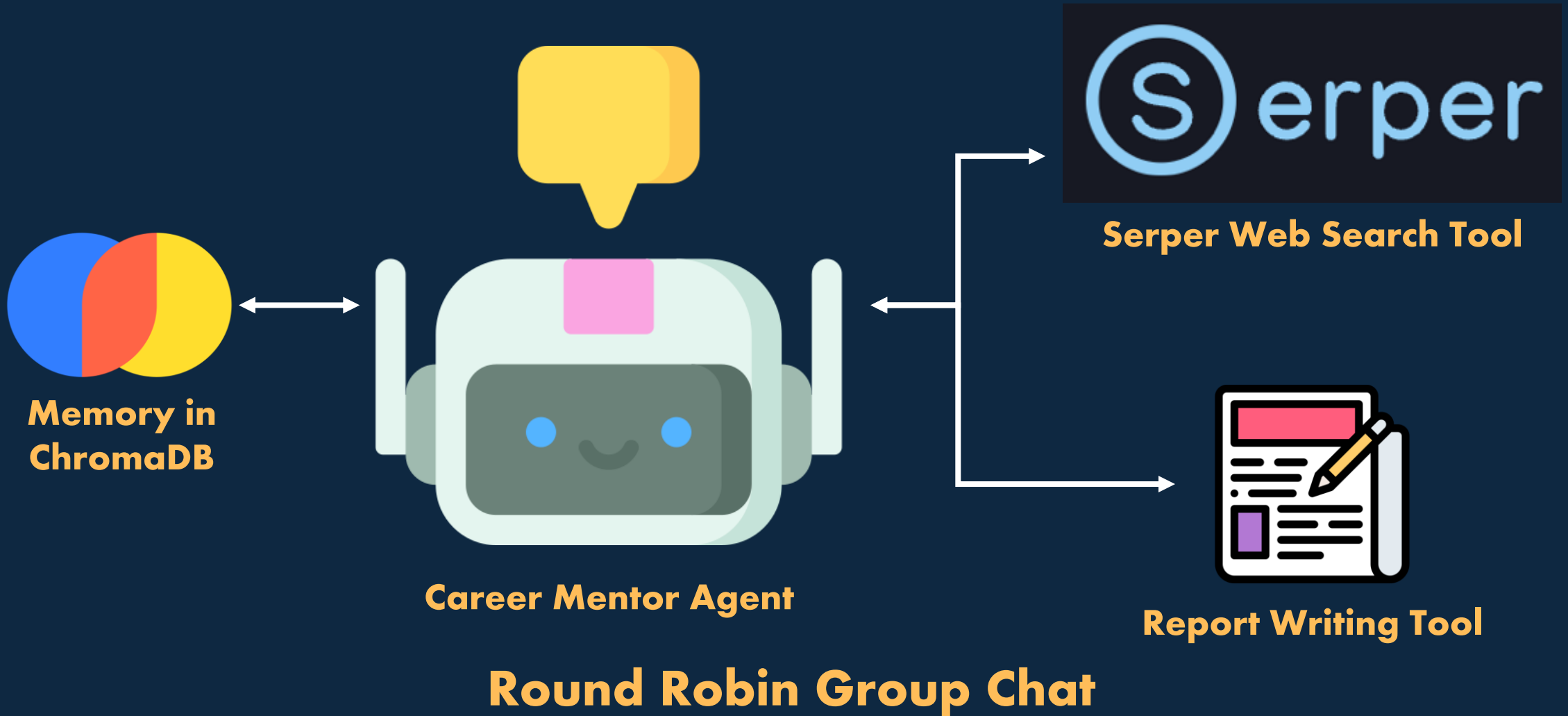


Single Agent Team



Round Robin Group Chat

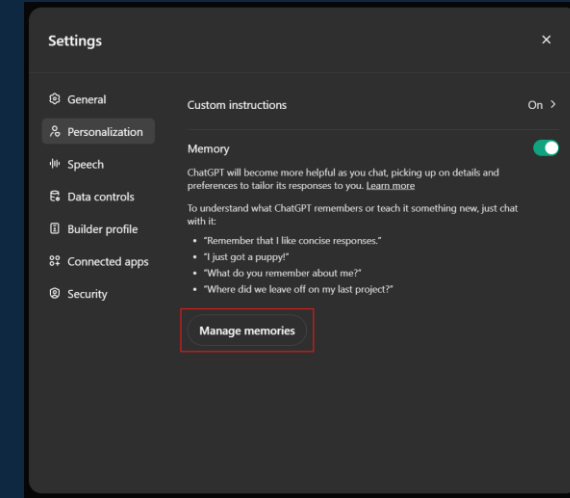
Single Agent Team with Memory



Single Agent Team with Memory

Why Memory?

- It may be valuable to maintain a store of useful facts that can be intelligently added to the context of the agent just before a specific step.
- It allows agents to build knowledge incrementally, reference past interactions appropriately, and maintain contextual awareness across multiple sessions.

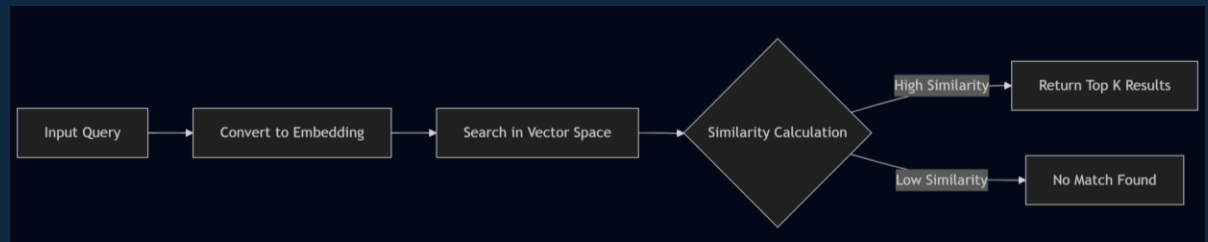


Memory in ChatGPT

Vector DB for Memory

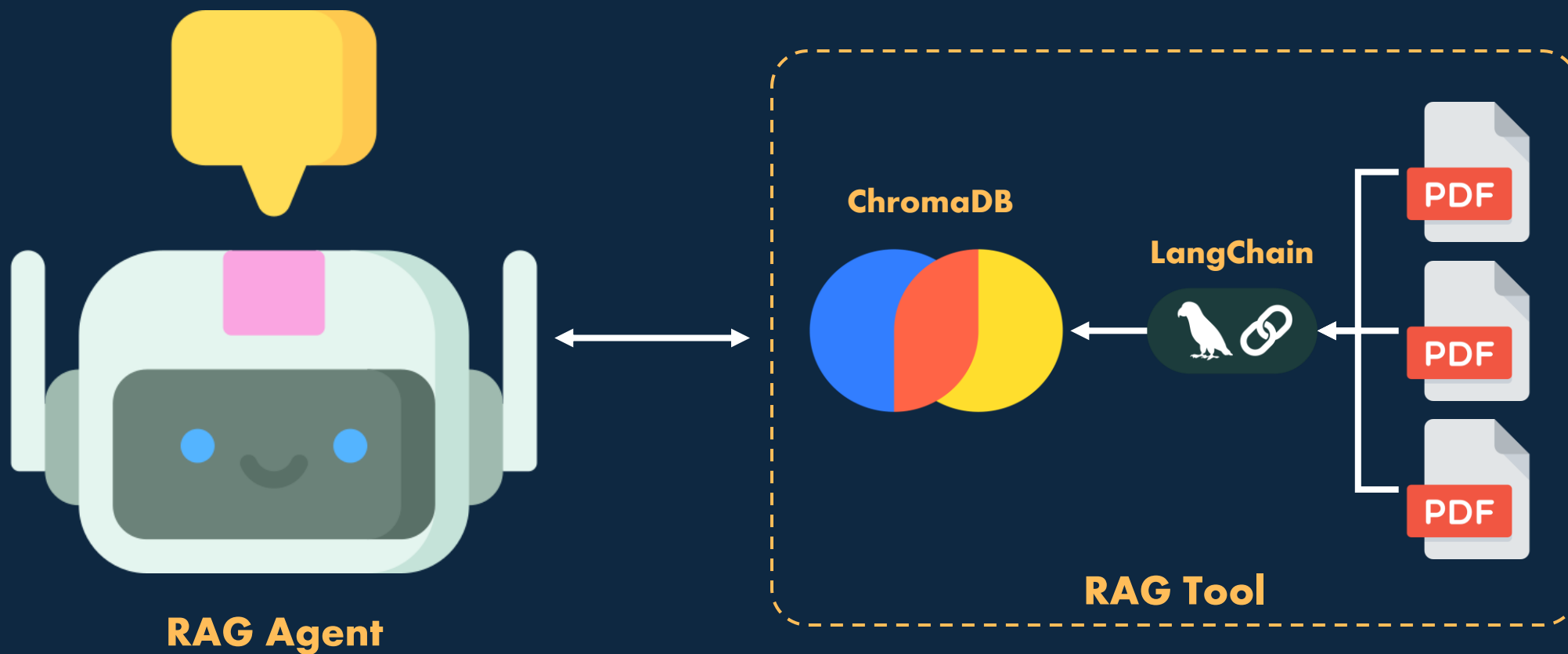
Vector store (e.g.: ChromaDB) memory provides:

- **Persistent storage** beyond individual sessions
- **Semantic search** capabilities for finding relevant past information
- **Efficient storage and retrieval** of large amounts of information



Semantic Retrieval from VectorDB

Single Agent Team with RAG



Round Robin Group Chat

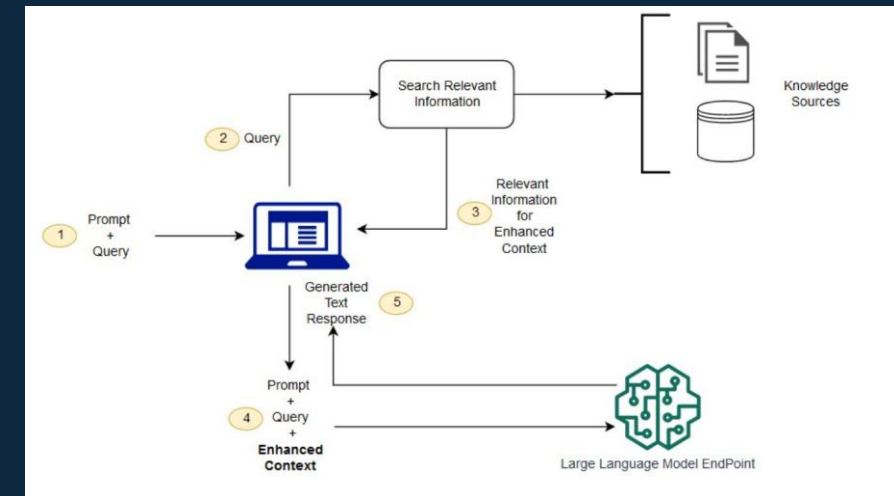
Single Agent Team with RAG

What is RAG?

- **Retrieval-Augmented Generation (RAG) is a powerful approach that enhances LLM outputs by retrieving relevant information from external knowledge bases**
- **The RAG process involves:**
 - **Converting user queries into vector embeddings**
 - **Searching a vector database for semantically similar documents**
 - **Retrieving relevant documents and incorporating them into the prompt context**
 - **Generating responses that leverage both the model's general knowledge and the retrieved information**

Why RAG using LangChain?

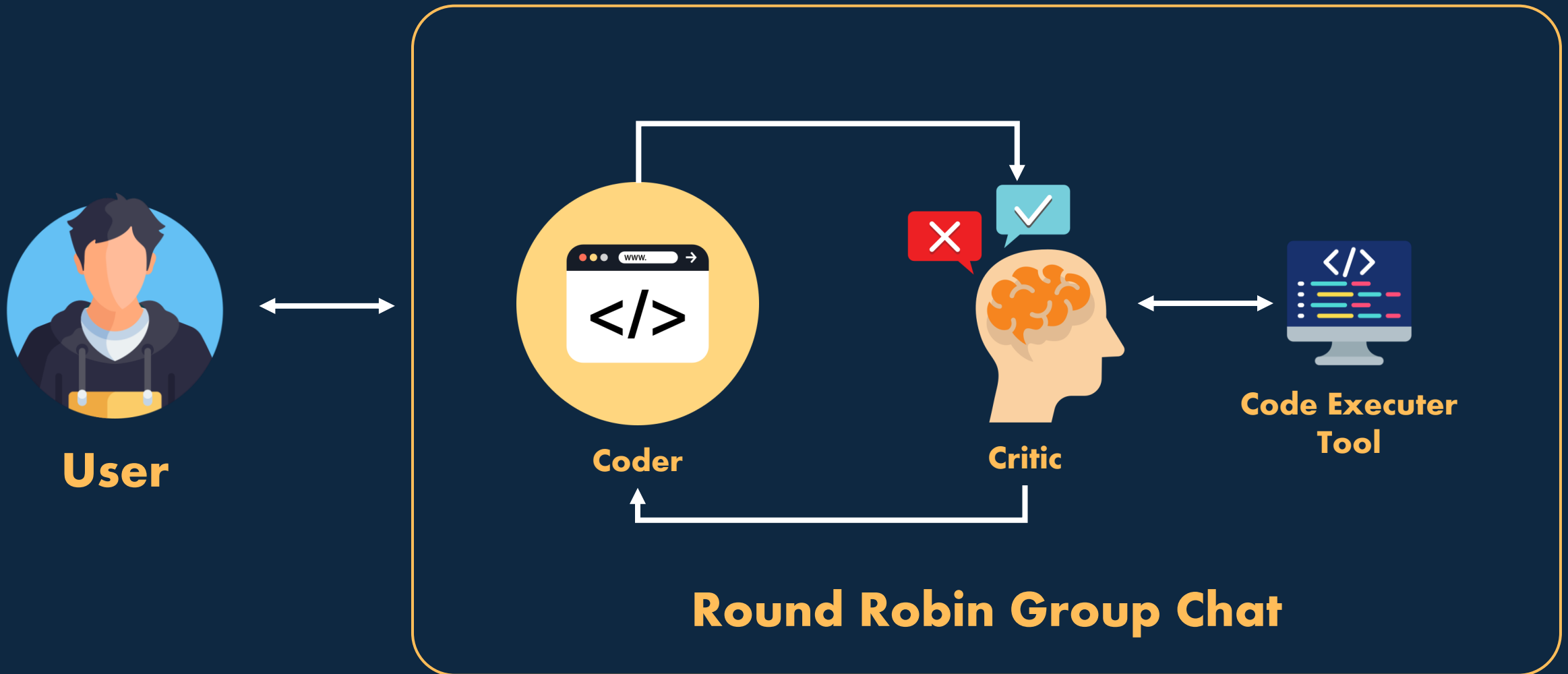
- **AutoGen does NOT have built-in RAG Tool**
- **AutoGen allows tools built with LangChain to be adapted**
- **LangChain offers variety of:**
 - **Document Loaders**
 - **Chunking Strategies**
 - **Integration with ChromaDB**



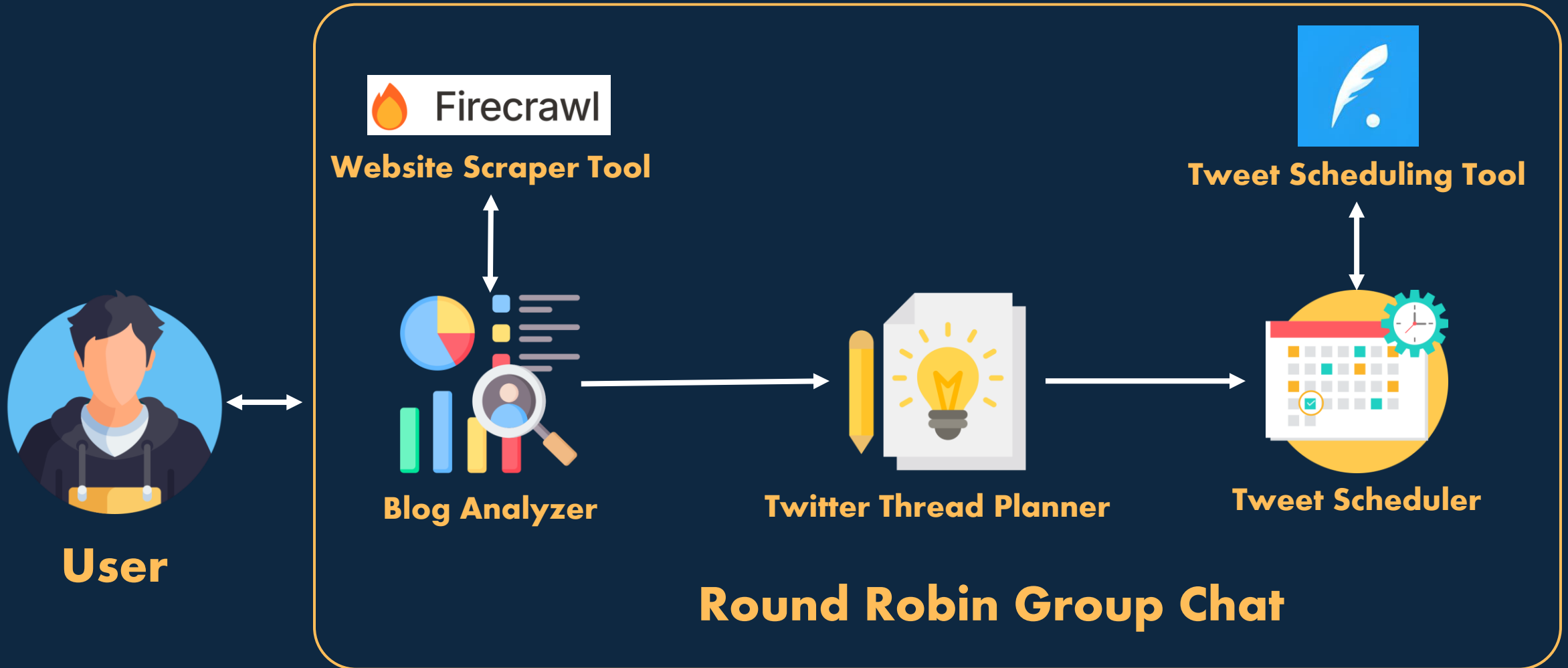


Building Multi-Agent Systems

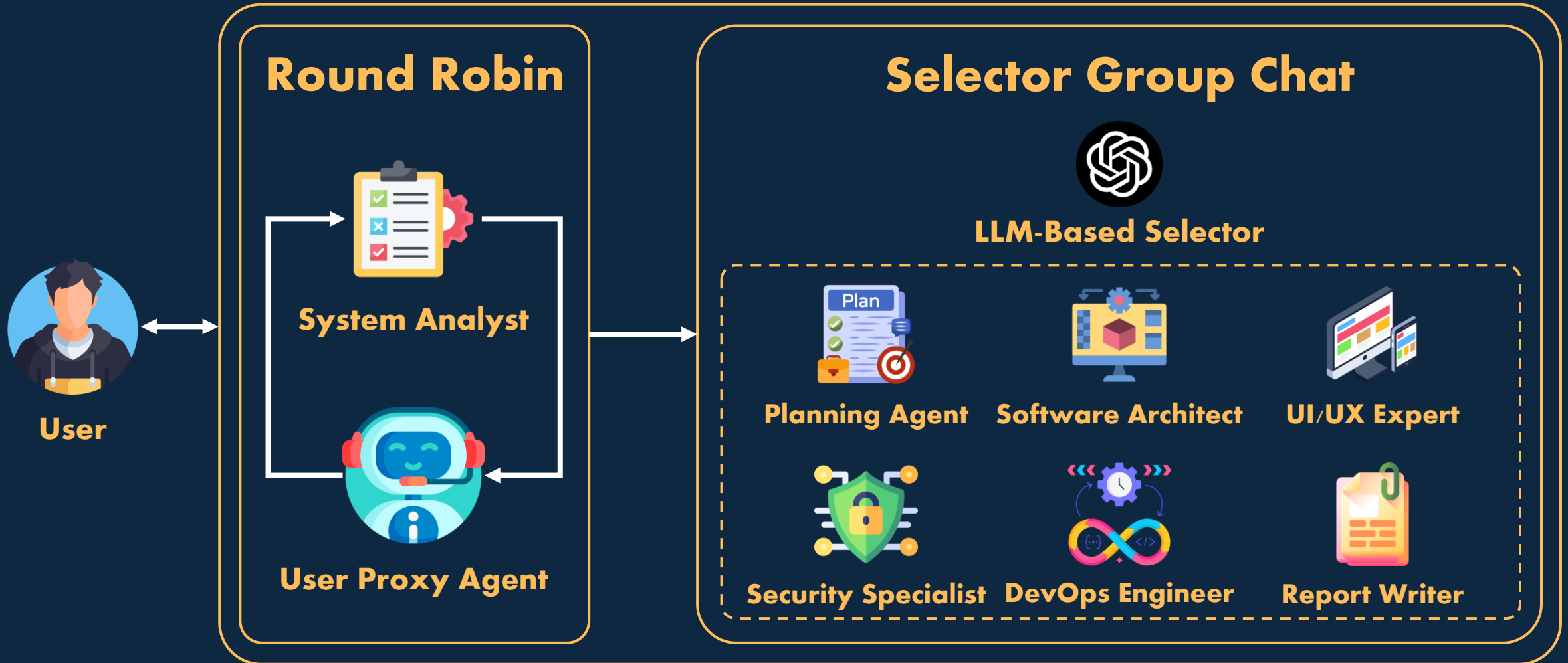
Coder-Reviewer Agents – Reflection Pattern



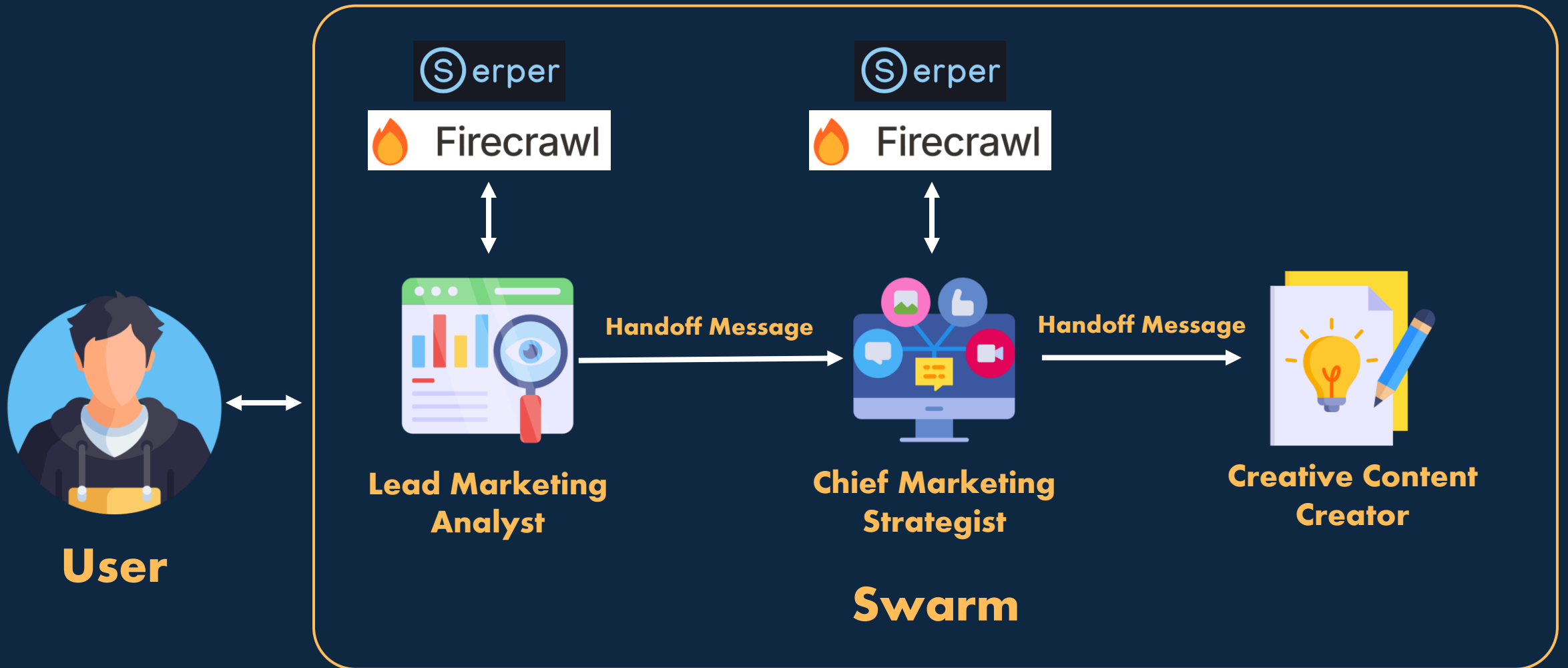
Blog-to-Twitter Thread Scheduler – Sequential



System Design Agent – Human-in-the-Loop + Planning



Marketing Campaign Creator – Swarm



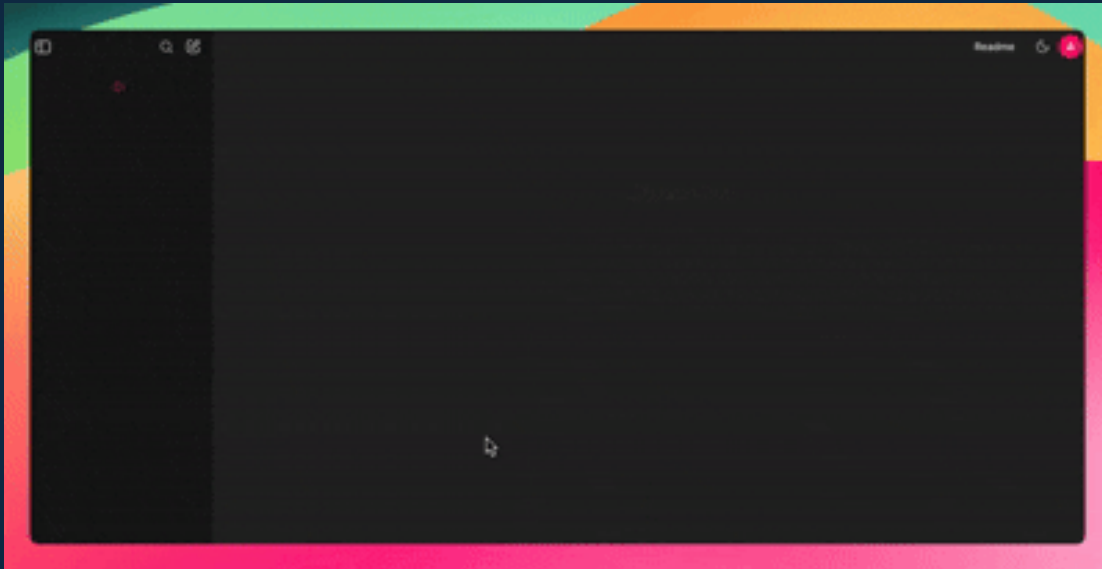
Taking AI Agents to Production

The background of the slide is a dark, atmospheric server room. In the center, a humanoid robot with a metallic body and a circular headpiece is seated at a desk, looking at a large, glowing digital display. Above the robot, a large, stylized blue cloud contains several server racks and data visualizations. The room is filled with rows of server racks on both sides, with various lights and indicators visible. The overall color palette is dark blue and black, with bright yellow and white highlights from the digital elements.

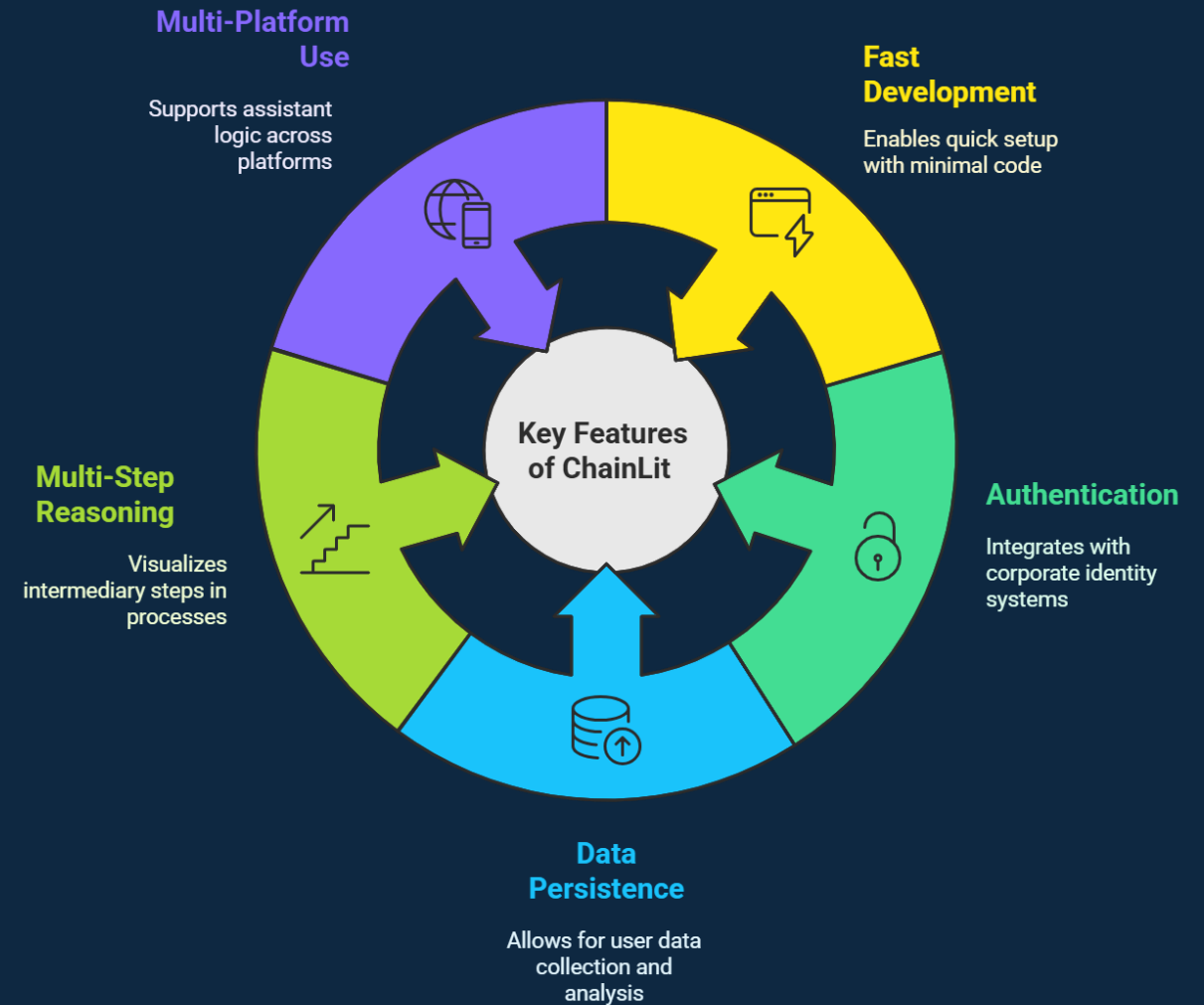
Putting a Face to Agentic Systems – using ChainLit

What is Chainlit?

An open-source Python package to build production-ready Conversational AI.



- [GitHub Repository](#)
- [Documentation](#)



Observability of AI Agents

Transparency

Tracing how inputs transform into outputs & revealing the internal reasoning mechanism

Interpretability

Translating complex AI decisions into human-understandable insights

Continuous Monitoring

Tracking performance, detecting biases & ensuring consistent and reliable AI behavior

Why Observability Matters?

- Enables trust
- Ensures reliability and efficiency
- Facilitates early detection of anomalies

Key Metrics to Track

- **Performance Metrics:** Response times, latency, and throughput
- **Resource Utilization:** LLM token usage, cost
- **Error Rates:** Frequency and types of errors encountered
- **User Interactions:** Engagement levels and satisfaction scores



How to Design Resilient AI Agents?

-  **What if an agent is stuck in a loop?**
 - How would you detect and resolve infinite loops or repetitive behavior?
-  **What if the agent does not respond within the allotted time?**
 - How would you handle timeouts and ensure graceful fallback?
-  **What if the agent loses internet connectivity mid-task?**
 - Can the agent recover or retry after a network failure?
-  **What if the agent receives incomplete or corrupted input?**
 - How would you validate and clean the input before processing?
-  **What if the agent generates sensitive or harmful content?**
 - What safeguards and monitoring mechanisms can you implement?

Time to Build Something Epic...

What Agentic AI System will you build? 🤔

Here are some thoughts to spark off your creative process...



AI-Powered Legal Bench

Contract Analyst, Legal Researcher, Negotiator, Compliance Agent, Reviewer



Stock Market Hedge Fund

Market Analyst, News Sentiment Analyzer, Strategist, Trader, Risk Manager



Book-Writing Crew

Plot Architect, Character Developer, Researcher, Writer, Editor

Quiz Time!!!



Scan the QR Code
Use the code 5984 7439 on menti.com

That's All Folks!

For More Interesting Stuff on AI , Data Science & the Latest Technologies...

